Austin-Chris Iwu

Statement of research intent

B.Eng. Electrical and Electronics Engineering, FUTO, 2025 · Second Class Upper, CGPA 4.09 / 5.00 · austinchrisiwu@gmail.com · austinchris.me

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I want to work on sensing and advice for smallholder farmers: how a cheap device in the field produces a measurement someone can act on, in a language they already speak, when the network is poor and a wrong recommendation costs a season.

That is the question I started in my final-year project, AgroSense360. I built an ESP32 and Arduino rover, a vision pipeline, and a React and Firebase dashboard that returns crop-health advice in local languages. The hardware, the model, and the interface were one system. What the project did not do, and what I want a lab for, is a proper study of whether the advice is right, and whether a farmer will use it.

The research I am proposing has three parts. First, sensor fusion under field noise: camera frames plus the cheap environmental sensors an ESP32 can actually carry, not a lab bench. Second, where the inference should run. A phone or the rover itself has to keep working when the uplink drops, so I care about what can stay on the device and what is worth sending. Third, evaluation with users. A labelled leaf dataset is not the same as a farmer deciding whether to spray. I want the study designed around that decision.

A manuscript on AgroSense360 is in preparation. I do not yet have a published journal paper, conference paper, or poster. The degree project is the research I have done. Around it I have two years of shipping production software at First Digit Communications (learning platforms, commerce, a school directory) and two hackathon wins in 2026: Earmark, Best Stablecoin Adoption at Celo Agents at Work, and Equilux, Wave 1 of the Midnight Buildathon. Those are not agriculture. They are the same habit: a constraint the user can check, built through to something that runs.

I am applying to work with a supervisor in embedded systems, computer vision, or agricultural engineering who wants a student who can build the apparatus and stay for the evaluation. If your group is on a different problem, I can rewrite this page around that problem. I would rather send you a statement about your lab than a general one.